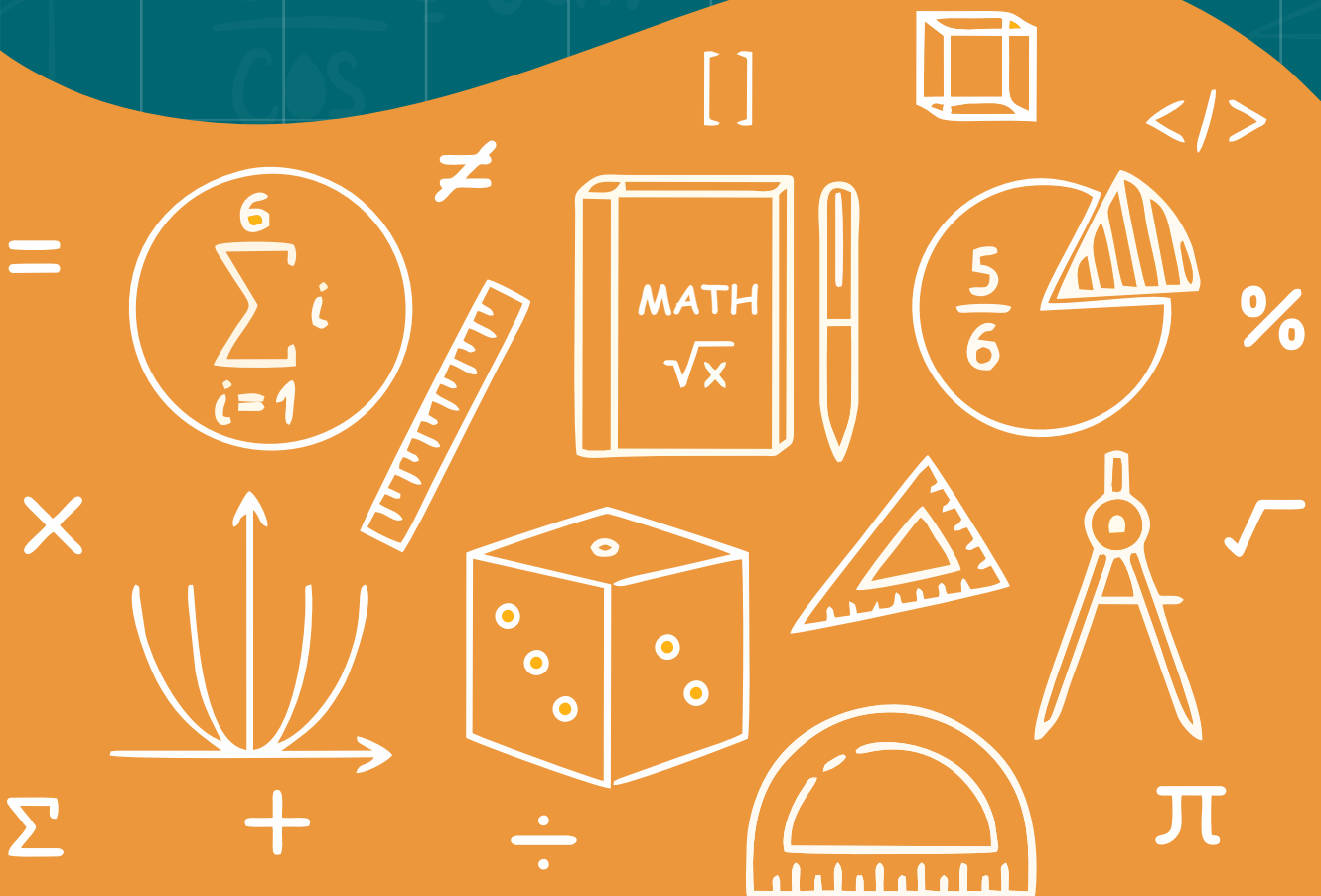


UNIT 2

WHOLE NUMBERS



NATURAL NUMBER

We use numbers '1, 2, 3, 4, 5...' for representing the quantity of objects in our daily life. These numbers 1, 2, 3, 4.... are called natural numbers.

The set of natural numbers is denoted by N.

i.e. $N = \{1, 2, 3, 4, 5, 6, \dots\}$

WHOLE NUMBER

When we include '0' in the set of natural numbers, we will get the set of whole numbers. '0, 1, 2, 3, 4...' are called whole numbers.

The set of whole numbers is denoted by W.

i.e. $W = \{0, 1, 2, 3, 4, 5, 6, 7, \dots\}$

Differentiate between natural and whole numbers:

The only difference between set of natural numbers and set of whole numbers is of number '0', '0' does not belong to the set of Natural numbers. All the other members of both the sets are the same.

- The smallest and the first natural number is 1.
- The first and the smallest whole number is '0'.
- The set of natural numbers is an infinite as we cannot count all natural numbers.
- The set of whole numbers is an infinite set.

Represent a given list of whole numbers:

Let us represent numbers 0, 1, 2, 3, 4, 5, ..., on a number line.

Steps:

- Draw a line 
- Mark a point to represent the first whole number '0' on the line.
- Mark other points at equal distance and name them 1, 2, 3, 4, 5, ... respectively as shown below:

Thus we get a number line as under:



Note

(i) On a number line, a number is greater than any number on its left. For example, $1 > 0$, $2 > 1$, and $3 > 2$, etc.

(ii) On a number line, a number is less than any number on its right. For example, $0 < 2$, $1 < 3$, $5 < 12$, etc.

Represent whole numbers $<$ or $>$ a given whole number on a number line:

Example.

Represent the following on number line.

(i) Whole numbers less than 6

(ii) Whole numbers greater than 4

Solution:

(i) Whole numbers less than 6

Number less than 6 means numbers to the left side of 6.



Dark shaded dots represent the required whole numbers.
So, 0, 1, 2, 3, 4, 5 are the whole numbers less than 6.

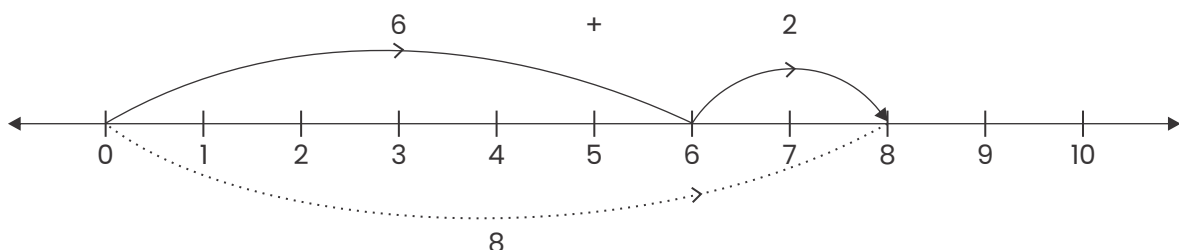
(ii) Whole numbers greater than 4



Dark shaded dots represent the required whole numbers.
So, 5, 6, 7, ... are the required whole numbers.

Example 1. Find the sum of 6 and 2 by using number line.

Solution:



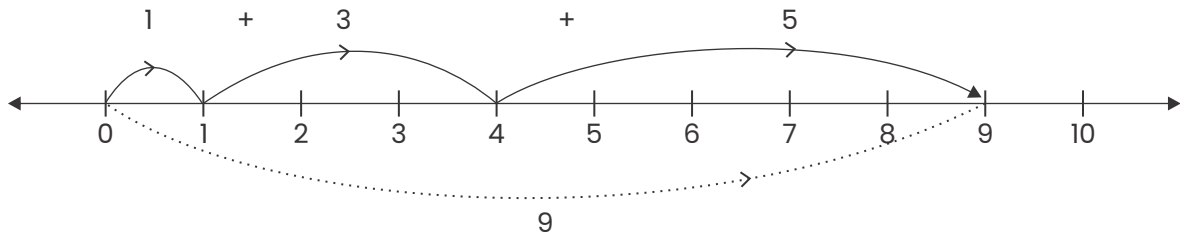
Its procedure is described below:

Start from 0, and move 6 units on the right. Again, starting from 6 move 2 units on the right. We reach at number 8.

Hence, $6+2=8$

Example 2. With the help of a number line, find the sum of $1 + 3 + 5$.

Solution:



Therefore, $1+3+5=9$

EXERCISE 2.1

Q1. Write, if possible:

(i) The smallest natural number.

(ii) The smallest whole number.

(iii) The largest natural number.

(iv) The largest whole number.

(v) Write first ten whole numbers.

(vi) Write first ten natural numbers.

Q2. Represent on number line.

- (i) Whole numbers greater than 6.
- (ii) Whole numbers greater than 2.
- (iii) Whole numbers less than 5.
- (iv) Whole numbers greater than 7.
- (v) Whole numbers equal to 10.

Q3. Find sum of the following whole numbers by using number line.

- (i) 3 and 6
- (ii) 7 and 9
- (iii) 4, 6 and 9
- (iv) 2 and 5
- (v) 3, 5 and 7

ADDITION AND SUBTRACTION OF WHOLE NUMBERS**Add and subtract two given whole numbers:**

We already know how to add and subtract two natural numbers. Same methods are applied for whole numbers.

Example 1: Add 526 and 214.

Solution:

	5	2	6
+	2	1	4
	7	4	0

Example 2: Subtract 233 from 467

Solution:

	4	6	7
-	2	3	3
	2	3	4

Example 3: Ali earns Rs 42,750 in a month, and his sister earns Rs 28,300. Their combined monthly expenditure is Rs 55,000. What is their total monthly income and how much do they save?

Solution:

Adding the monthly income of Ali and his sister:

Monthly income of Ravi = 42,750 rupees

Monthly income of his sister = 28,300 rupees

Total of monthly income: $42,750 + 28,300 = 71,050$ rupees

Now, in order to find monthly saving, we subtract monthly expenditure from their total income.

Monthly income: 71,050 rupees

Monthly expenditure: 55,000 rupees

Monthly saving: $71,050 - 55,000 = 16,050$ rupees

Therefore, their total monthly income is Rs 71,050 and their monthly saving is Rs 16,050.

Example 4: Find the sum of the smallest six-digit number and the largest five-digit number.

Solution:

Smallest six-digit number:	100000
Largest five-digit number:	+ 99999
	199,999

So, the required sum is 199,999.

**COMMUTATIVE AND ASSOCIATIVE LAW (UNDER ADDITION)
OF WHOLE NUMBERS.****(a) Commutative law under addition.**

$$9 + 30 = 30 + 9, \text{ and } 100 + 40 = 40 + 100$$

It shows that, for any order of two given whole numbers, their addition gives us the same sum. This is known as the commutative law of whole numbers under addition.

Thus, commutative law under addition is:

The sum of two whole numbers, in any order, is always same.

Example: Verify the commutative law under addition for whole numbers 57 and 43.

Solution:

By Commutative Law under addition, $57 + 43 = 43 + 57$

$$\text{LHS} = 57 + 43$$

$$= 100$$

or

$$\begin{array}{r} 57 \\ + 43 \\ \hline 100 \end{array}$$

$$\text{RHS} = 43 + 57$$

$$= 100$$

or

$$\begin{array}{r} 43 \\ + 57 \\ \hline 100 \end{array}$$

(b) Associative law under addition.

We can add only two whole numbers at a time. So, we can get the sum of three whole numbers, say 3, 7, and 12 by two ways:

$$(3 + 7) + 12$$

$$= 10 + 12$$

$$= 22$$

or

$$3 + (7 + 12)$$

$$= 3 + 19$$

$$= 22$$

Therefore we can say:

$$(3 + 7) + 12 = 3 + (7 + 12)$$

Hence, three whole numbers added in any order give the same result. This is known as associative law under addition.

Thus, Associative law under addition is:

The sum of three whole numbers, in any order, is always same.

EXERCISE 2.2

Q1. What is the sum of the smallest number of five digits and the largest number of four digits?

Q2. A school has 1,200 students. Out of them, 450 are in the sports club, and 380 are in the science club. The rest are not part of either club. How many students are not in any of these clubs?

Q3. Sara deposited Rs 30,000 in her savings account. After two months, she withdrew Rs 7,500. How much money remains in her account?

Q4. Rayan had Rs 50,000. He spent Rs 8,000 on groceries, Rs 12,000 on rent, and Rs 5,500 on utilities. How much money is left with him?

Q5. Verify commutative law under addition for the following.

(i) 12345 and 67890

(ii) 54321 and 98765

(iii) 234567 and 890123

(iv) 1001 and 2002

Q6. Verify associative law under addition for the following.

(i) 1250, 340, and 5678

(ii) 4321, 987, and 6543

(iii) 102, 205, and 3001

(iv) 8750, 129, and 40000

Q7. Fill in the blanks to make the following statements true.

(i) $1000 + 250 = \underline{\hspace{2cm}} + 1000$

(ii) $45678 + 0 = \underline{\hspace{2cm}}$

(iii) $300 + 0 = 0 + \underline{\hspace{2cm}}$

(iv) $560 + (120 + 80) = (560 + \underline{\hspace{2cm}}) + 80$

(v) $750 + 150 = 150 + 750 = \underline{\hspace{2cm}}$

MULTIPLICATION AND DIVISION OF WHOLE NUMBERS

(a) Multiplication:

The method of multiplying two whole numbers is same as the method of multiplication of natural numbers.

The product of two whole numbers is always a whole number.

Consider the following multiplications:

$$3 \times 5 = 15, \quad 5 \times 5 = 25, \quad 0 \times 456 = 0$$

Example 1: Multiply 7747 by 696

Solution:

				7	7	4	7
				x	6	9	6
			4	6	4	8	2
	6	9	7	2	3	x	
4	6	4	8	2	x	x	
5	3	9	1	9	1	2	

Hence $7747 \times 696 = 5391912$

Example 2: Aisha bought 12 note books at the price of Rs 34 each. Find the total amount paid by Aisha.

Solution:

Number of notebooks:	12
Price of each notebook:	$\times 34$
	48
	$+ 36 \times$
	408

Hence the total amount is Rs 408.

(b) Division:

The method of dividing two whole numbers is same as the method of division of two natural numbers.

For example: If we divide 9 by 2, then the quotient is 4 and the remainder is 1.

$$\begin{array}{r} 4 \\ 2 \overline{) 9} \\ \underline{- 8} \\ 1 \end{array}$$

- The quotient of a whole number divided by 1 is the number itself.
- The quotient of zero divided by a non-zero whole number is zero.

Let us consider division of some other whole numbers along with their quotients and remainders.

Dividend	Divisor	Quotient	Remainder
9	3	3	0
7	5	1	2
14	8	1	6
56	7	8	0

Example 2: The cost of 15 books is Rs 600. Find the cost of one book.

Solution:

In order to find the cost of one book we divide the total cost Rs 600 by number of books, i.e. 15.

$$\begin{array}{r} 40 \\ 15 \overline{) 600} \\ \underline{- 600} \\ \text{xxx} \end{array}$$

So, the cost of one book is Rs 40.

EXERCISE 2.3

Q1. Find the products of the following whole numbers.

(i) 563×67

(ii) 647×763

(iii) 5336×837

(iv) 674×384

Q2. Find the product.

$$\begin{array}{r} 2032 \\ \times 93 \\ \hline \end{array}$$

$$\begin{array}{r} 9836 \\ \times 22 \\ \hline \end{array}$$

$$\begin{array}{r} 7784 \\ \times 91 \\ \hline \end{array}$$

$$\begin{array}{r} 4049 \\ \times 61 \\ \hline \end{array}$$

$$\begin{array}{r} 3699 \\ \times 54 \\ \hline \end{array}$$

$$\begin{array}{r} 7514 \\ \times 14 \\ \hline \end{array}$$

Q3. Divide and find the quotient and remainder.

(i) $4321 \div 65$

(ii) $85000 \div 4000$

(iii) $123456 \div 789$

(iv) $65432 \div 250$

Q4. Solve the following.

(i) $75 \overline{)56,569}$

(ii) $68 \overline{)56,855}$

(iii) $42 \overline{)51,860}$

(iv) $36 \overline{)97,722}$

(v) $92 \overline{)74,186}$

(vi) $25 \overline{)6,231}$

WORD PROBLEMS

Q1. A farmer has 240 apples that he wants to pack into boxes. If each box can hold 12 apples, how many boxes will he need?

Q2. A bookshop has 75 novels. If each novel costs Rs 450, find the total cost of all the novels.

Q3. Find the smallest 5-digit number which is exactly divisible by 50.

Q4. Subtract the smallest number of three digits from the largest number of four digits. What is the result?

COMMUTATIVE AND ASSOCIATIVE LAW (UNDER MULTIPLICATION) OF WHOLE NUMBERS:

(a) Commutative law under multiplication:

Two whole numbers multiplied in any order gives equal product. This is known as commutative law under multiplication.

The product of two whole numbers, in any order, is always same.

(b) Associative law under multiplication:

Three whole numbers multiplied in any order gives equal product. This is the associative law under multiplication. Thus, associative law under multiplication is:

The product of three whole numbers, in any order, is always same.

Distributive law of multiplication over addition and subtraction :

Let us demonstrate the distributive law of multiplication over addition with the help of example $5 \times (10 + 8) = 5 \times 10 + 5 \times 8$

$$\begin{aligned}\text{L.H.S} &= 5 (10 + 8) \\ &= 5 (18) \\ &= 90\end{aligned}$$

$$\begin{aligned}\text{R.H.S} &= 5 \times 10 + 5 \times 8 \\ &= 50 + 40 \\ &= 90\end{aligned}$$

$$\therefore \text{L.H.S} = \text{R.H.S}$$

$$\therefore 5 (10 + 8) = 5 \times 10 + 5 \times 8$$

Let us demonstrate of Distributive law of multiplication over subtraction with help of example $5 (20 - 2) = 5 \times 20 - 5 \times 2$

$$\begin{aligned}\text{L.H.S} &= 5 (20 - 2) \\ &= 5 (18) \\ &= 90\end{aligned}$$

$$\begin{aligned}\text{R.H.S} &= 5 \times 20 - 5 \times 2 \\ &= 100 - 10 \\ &= 90\end{aligned}$$

$$\therefore \text{L.H.S} = \text{R.H.S}$$

$$\therefore 5 (20 - 2) = 5 \times 20 - 5 \times 2$$

EXERCISE 2.4**Q1. Fill in the blanks.**

(i) $7 \times 0 =$ _____

(ii) $9 \times 1 =$ _____

(iii) $3 \times 4 = 4 \times$ _____

(iv) $2 \times (3 \times 5) = (2 \times 3) \times$ _____

(v) $4 \times (\text{_____} \times 6) = (\text{_____} \times 5) \times 6$

(vi) $300 \times (150 + \text{_____}) = 300 \times 150 + 300 \times 36$

(vii) $200 \times (100 - 50) = \text{_____} \times 100 - \text{_____} \times 50$

Q2. Write True or False.

(i) $123 \times 456 = 123 + 456$

(ii) $50 - (30 - 10) = (50 - 30) - 10$

(iii) $12(3 + 4) = 12 \times 3 + 12 \times 4$

(iv) $10(6 \times 5) = (10 \times 6) \times 5$

Q3. Name and verify the law used in each of the following.

(i) $8 \times 5 = 5 \times 8$

(ii) $3 \times (2 \times 4) = (3 \times 2) \times 4$

(iii) $6 \times (10 - 2) = 6 \times 10 - 6 \times 2$

(iv) $5 \times (9 + 1) = 5 \times 9 + 5 \times 1$

Q4. The distributive property states that:

$$a \times (b + c) = (a \times b) + (a \times c)$$

Example: 10×21 **Solutions:**

1. 10×21

$$\begin{aligned} 10 \times (20 + 1) &= (10 \times 20) + (10 \times 1) \\ &= 200 + 10 = 210 \end{aligned}$$

Now, let's apply this property to solve the following equations:

(ii) 3×12

(iii) 8×26

(iv) 3×28

(v) 4×23

Q5. Use the associative property to fill the missing values.

(i) (_____ x 3) x 8 = (_____ x 3) x 6

(ii) (76 x _____) x 9 = (_____ x 3) x 76

(iii) (_____ x 87) x 17 = 17 x (_____ x 4)

(iv) _____ (14 x 90) = 90 x (_____ x 9)

(v) (89 x _____) x 8 = 8 x (_____ x 43)

(vi) (_____ x 7) x 6 = (_____ x 5) x 6

Q6. Use the commutative property to fill the missing values.

1. _____ x 3 = 3 x 43

2. _____ x 5 = 5 x 8

3. _____ x 5 = 5 x 6

4. 3 x _____ x 9 x 3

5. 6 x _____ = 2 x 6

6. 6 x 82 = _____ x 6

7. 6 x 14 = 14 x _____

8. 2 x 88 = _____ x 2

Q7. Verify the Commutative Law under Addition for 125 and 875.

Q8. Verify the Associative Law under Multiplication for 15, 4, and 10.